SUMMARY · 27 OCTOBER 2026

CFD of the textured-blade turbine

What 263 simulations show about blade roughness, and what to test next

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In short. In the simulations, rough blades can increase the turbine's torque. The gain comes mostly from the moment a blade moves edge-on into the wind, not from the cup-versus-back drag that we expected to matter. With very rough blades (400 µm) the simulated torque rose by about a third, the same direction as our wind-tunnel gains. With roughness closer to our printed blades (100 µm) there was no clear change. The simulations are two-dimensional and sensitive to their settings, so they cannot confirm the mechanism on their own: a direct force measurement on a blade is the next step.

1 Answers to the questions from our 7 October meeting

Question	Answer
Lift and drag coefficients of the blade (theory and experiment)	**From the CFD** (smooth blade, 23 m/s, 2D, chord 48 mm): cup facing the wind, drag 1.8–2.0 or about 3.2, depending on which of two flow patterns forms, with lift near zero; edge-on, drag 0.35–0.42 and lift 0.7–1.1; back facing the wind, drag 1.5–1.7, falling to 1.1–1.2 with a finer time step. For comparison, published values for a thin half-tube are 2.3 (cup facing) and 1.2 (back facing). **By experiment**: not yet measured; the force test is designed (Section 6).
A CFD model in OpenFOAM	Done: 263 two-dimensional simulations of the blade and the rotor, summarised in the rest of this report.
Cost and time per blade; a shorter blade?	Print Lab slicer estimates, per blade: FS 0.10, 7 h 40 min, 39 g, $4.61; FS 0.20, 8 h 41 min, 41 g, $5.15 (both with a 0.2 mm nozzle); FS 0.80, 15 h 54 min, 72 g, $9.38 (with a 0.4 mm nozzle). The Print Lab charges $0.50 per print hour plus material. A shorter blade saves time and material roughly in proportion, but it changes the rotor, so it would need its own Plain set for comparison.
A rougher set ("320 µm", 0.805 mm fuzzy skin)	The Print Lab has the print settings ready. We propose printing FS 0.40 and FS 0.80 together with a fresh Plain set and repeats of FS 0.10 and FS 0.20, all with the same settings. "Ra 320" is a label: FS 0.20 ("Ra 80") measured Ra 10 µm, so we will scan the new blades.
Is there a best roughness?	Not known yet. In the CFD, very rough blades help the rotor, but realistic roughness gives no clear change. The new print round (five levels, two prints each, tested in mixed order) is designed to find it.
What the Sandia report predicts for us	Sandia (SAND2017-11288) tested lift-driven airfoils at much higher Reynolds numbers, where roughness lowers lift and raises drag (4.9–6.8 % energy loss at 200 µm). Our rotor is drag-driven at low Reynolds numbers, where roughness can make the boundary layer turbulent earlier, so Sandia does not predict a loss for us. Its roughness criterion suggests FS 0.20 reaches that point at our higher wind speeds, and that rougher blades should start gaining at lower wind speeds and then level off. The FS 0.40 and FS 0.80 tests can check this.

2 Why we ran these simulations

On 1 October, the FS 0.10 and FS 0.20 textured blades produced 12.8 % and 16.2 % more peak electrical power than the Plain blades (averaged over 10.2–38 m/s). We used OpenFOAM to ask two questions: how does roughness change the forces on a single blade, and how does it change the torque of the whole rotor?

3 What we simulated

All runs are two-dimensional slices through the middle of the blade, at 23 m/s unless stated, run on URI's Unity computing cluster.

Model	Cases	What was varied
Single blade	194	Angle to the wind (all the way round), wind speed (10.2, 23, 38 m/s), roughness height (smooth to 400 µm), and the printed texture shape itself
Whole rotor (3 blades)	69	Tip-speed ratio (blade-tip speed divided by wind speed), wind speed, roughness height, mesh resolution

Roughness enters the model as a uniform "sand-grain" height. Our blades' measured texture height is about 84 µm to 97 µm for FS 0.20, so the 100 µm cases are the closest to the real blades; 400 µm is a deliberately rough upper case.

4 Results

4.1 On a single blade, roughness mostly changes the edge-on forces

When the blade faces the wind cup-first or back-first, 400 µm roughness changes the drag by only about 6 %. When the blade is edge-on, the same roughness raises the drag by 45 % and the lift by 67 % (Fig. 1). At 100 µm all changes are small (10 % or less).

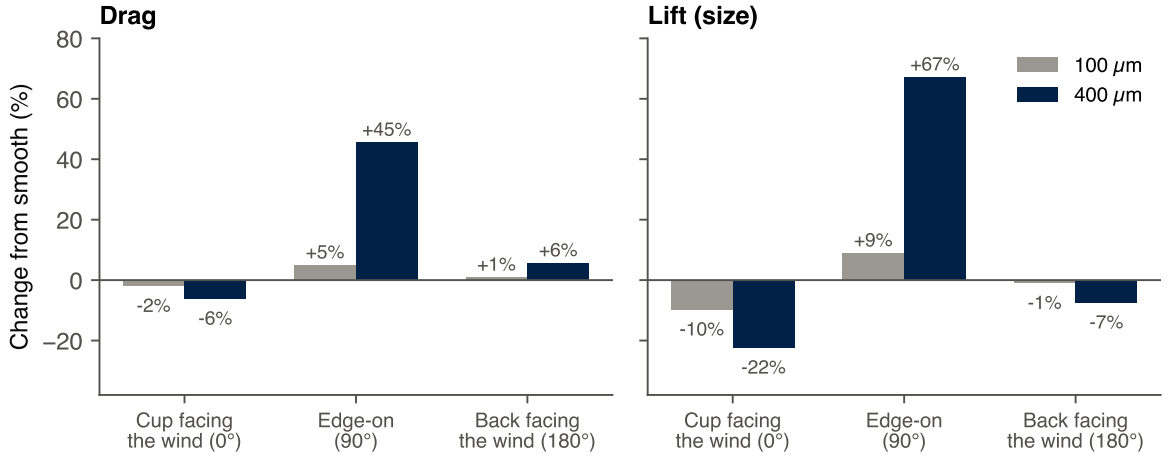


Figure 1. Change in drag and lift on a single blade from roughness, compared with a smooth blade, for three orientations (23 m/s).

4.2 On the rotor, heavy roughness raises the torque

With 400 µm roughness, the rotor's torque rose by 27 % at a tip-speed ratio of 0.10 and by 38 % at 0.12 (Fig. 2). Our turbine reached peak power near a tip-speed ratio of 0.12–0.13 in the wind tunnel. At 100 µm the changes (+8 % and −19 %) are within the simulations' noise, so we cannot say whether realistic roughness helps or hurts in this model.

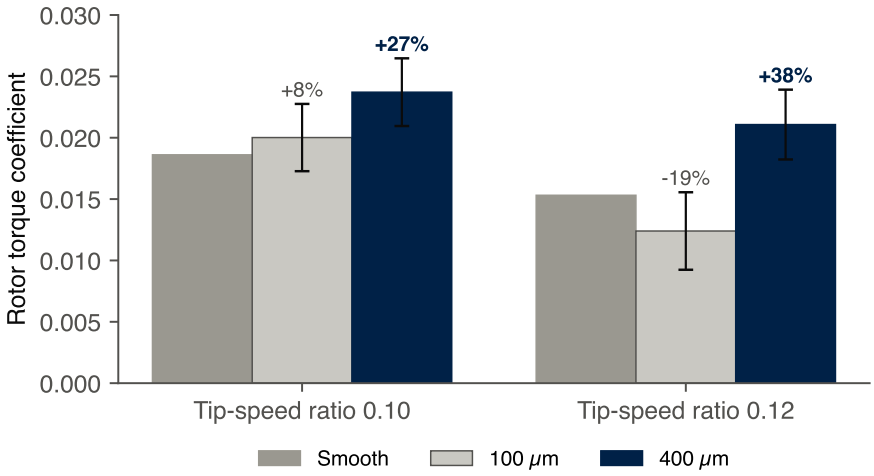


Figure 2. Rotor torque with smooth and rough blades at two tip-speed ratios (23 m/s, averaged over 45 revolutions). Error bars show the simulations' noise; a change is clear when it is larger than the bar.

4.3 The gain comes from the blade moving edge-on with the wind

Splitting the turn into positions shows where the extra torque comes from (Fig. 3). The blade moving edge-on with the wind contributes the most (+49 % of the smooth rotor's torque). Some is lost when the blade moves edge-on against the wind (−26 %). This matches the single-blade result: the edge-on forces are the ones roughness changes most. Our earlier, simpler estimate (13 October) looked only at the cup-facing and back-facing positions, which is why it predicted a loss.

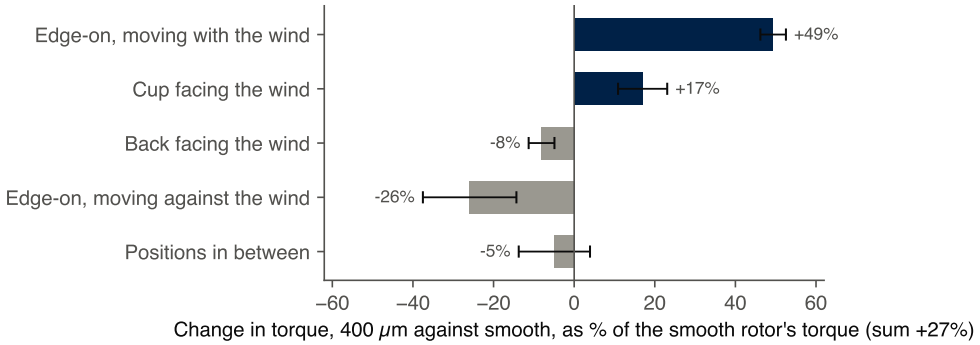


Figure 3. Where in each turn the 400 µm blades gain or lose torque, as a percentage of the smooth rotor's torque (tip-speed ratio 0.10). The parts add up to the +27 % of Fig. 2.

5 How much to trust this

- **The simulations are two-dimensional.** Flow around a curved blade is three-dimensional, and 2D simulations of this kind usually overstate drag. The results are best read as comparisons between cases rather than as absolute values.
- **The rotor result is promising but not yet firm.** The rotor's torque changes more between mesh resolutions than it does with roughness. We assumed that this affects smooth and rough runs the same way, but have not shown it.
- **The model's roughness is not our texture.** The printed fuzzy skin is not a uniform sand-grain surface, and we do not yet know which roughness height represents it.
- **One flow pattern is uncertain.** With the cup facing the wind, the simulated flow can settle into either of two stable patterns, one with nearly twice the drag of the other. Only a measurement can tell which one the real blade has.

6 Proposed next steps

1. **Measure lift and drag on single blades** (the "airfoil slice" test). An existing Plain blade and a textured blade, half in the wind, on a small two-load-cell balance under the tunnel floor, at the cup-facing, edge-on and back-facing angles. This tests the main simulation result directly. The design is ready; parts cost about $250–450.
2. **Test rougher blades.** Print FS 0.40 and FS 0.80 (the "320 µm" set) together with a fresh Plain set and repeats of FS 0.10 and FS 0.20, all with the same print settings, and test them in one session. I will ask the Print Lab for time and cost estimates.
3. **Record the rig details** the simulations had to assume: how the blades attach to the arms, which way the rotor turns, and the tunnel's test-section size.

Since the MCE labs now also use the tunnel, I am coordinating test times with Jim Byrnes and Rob D'Ambrosca.

Details, all figures and the scripts behind every number: `cfd/analysis/2026-10-15/` in the project repository.